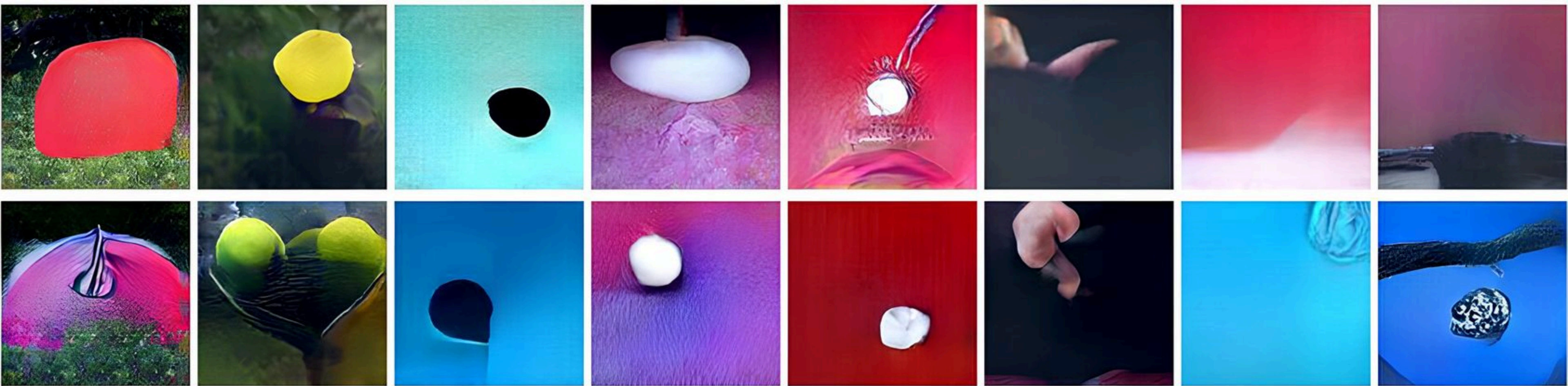




Emergent representations in AI latent spaces

An artistic and historical perspective

Paolo Berti



Venice Summer School
in Digital and Public Humanities

Two Schools

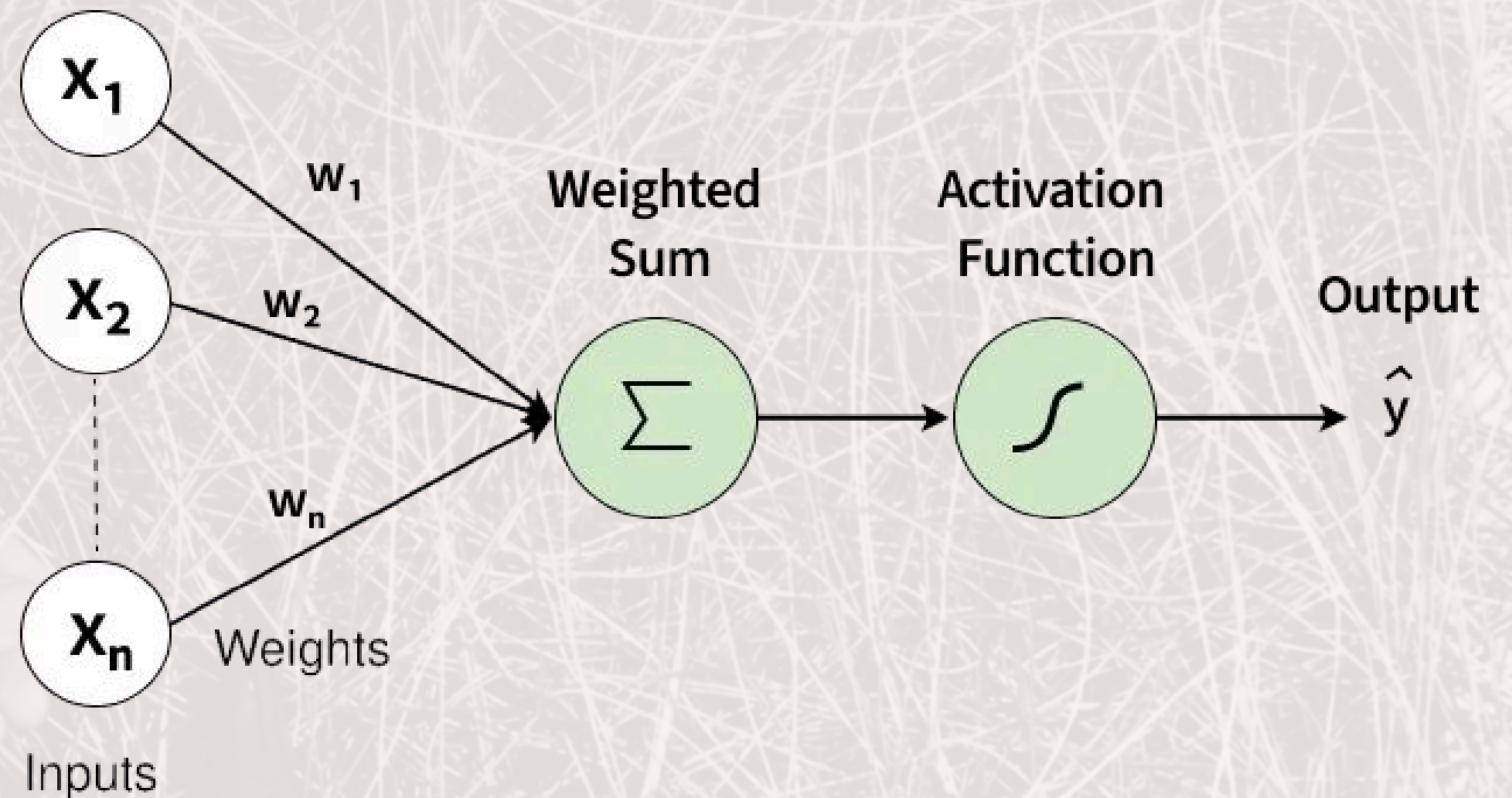
Diverging Approaches

The **symbolic** school focused on explicit rules, while the **connectionist** school emphasized learning from examples. Today, most advancements in AI are rooted in the connectionist approach, which imitates the workings of neurons.

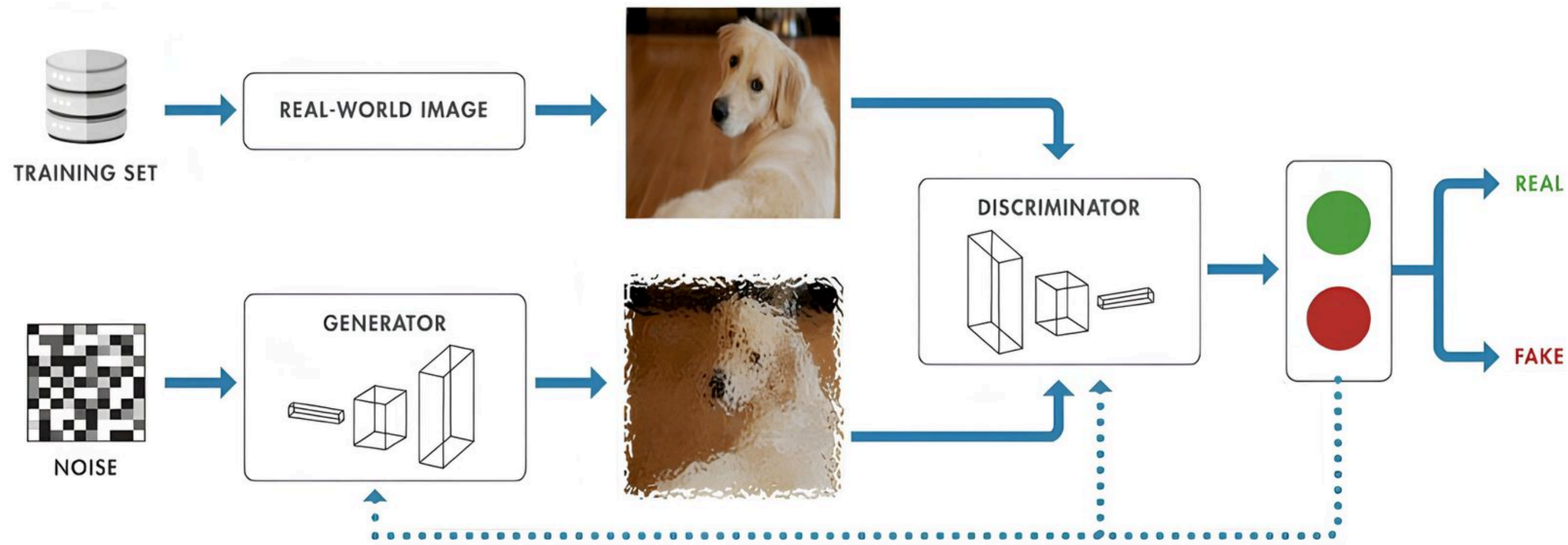


The Perceptron

Developed by Frank Rosenblatt, represents a **fundamental breakthrough** in artificial intelligence, mimicking biological neurons and enabling machines to learn from examples.



GENERATIVE ADVERSARIAL NETWORK (GAN)

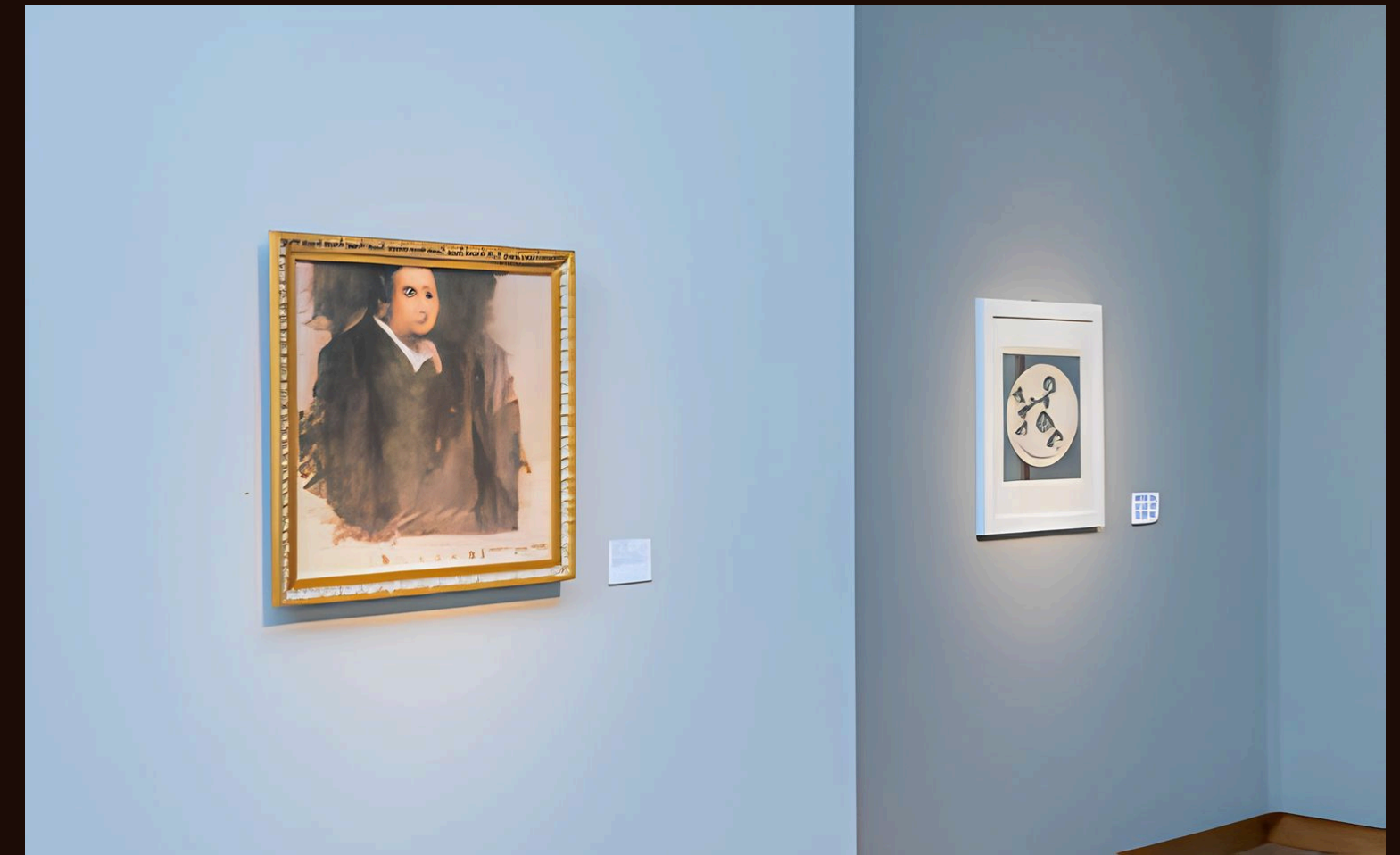
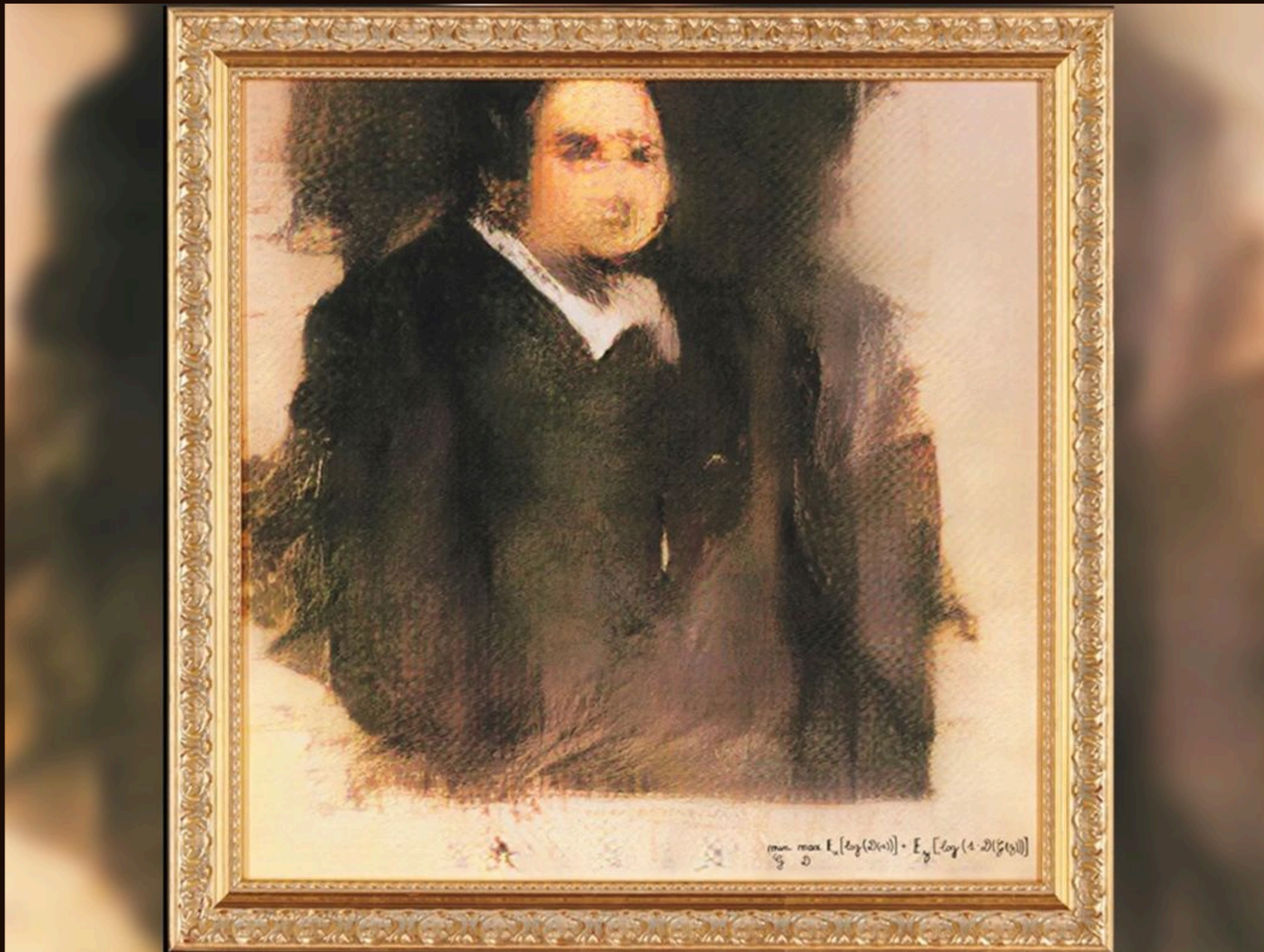


Generator

Creates images from random noise, simulating artistic creation

Discriminator

Evaluates generated images, determining authenticity and quality



Obvious, *Edmond de Belamy*, 2018



Robbie Barrat

@DrBarrat

Follow

left: the "AI generated" portrait Christie's is auctioning off right now

right: outputs from a neural network I trained and put online *over a year ago*.

Does anyone else care about this? Am I crazy for thinking that they really just used my network and are selling the results?



1:31 PM - 25 Oct 2018

1,641 Retweets 4,221 Likes



159 1.8K 4.2K



Twot your reply

+ TECH + AI + ENTERTAINMENT

How three French students used borrowed code to put the first AI portrait in Christie's

by + James Vincent

Oct 23, 2018, 3:34 PM GMT+2



0

Comments

Tweet by Robbie Barrat following the auction of *Portrait of Edmond Belamy*

UNITED STATES

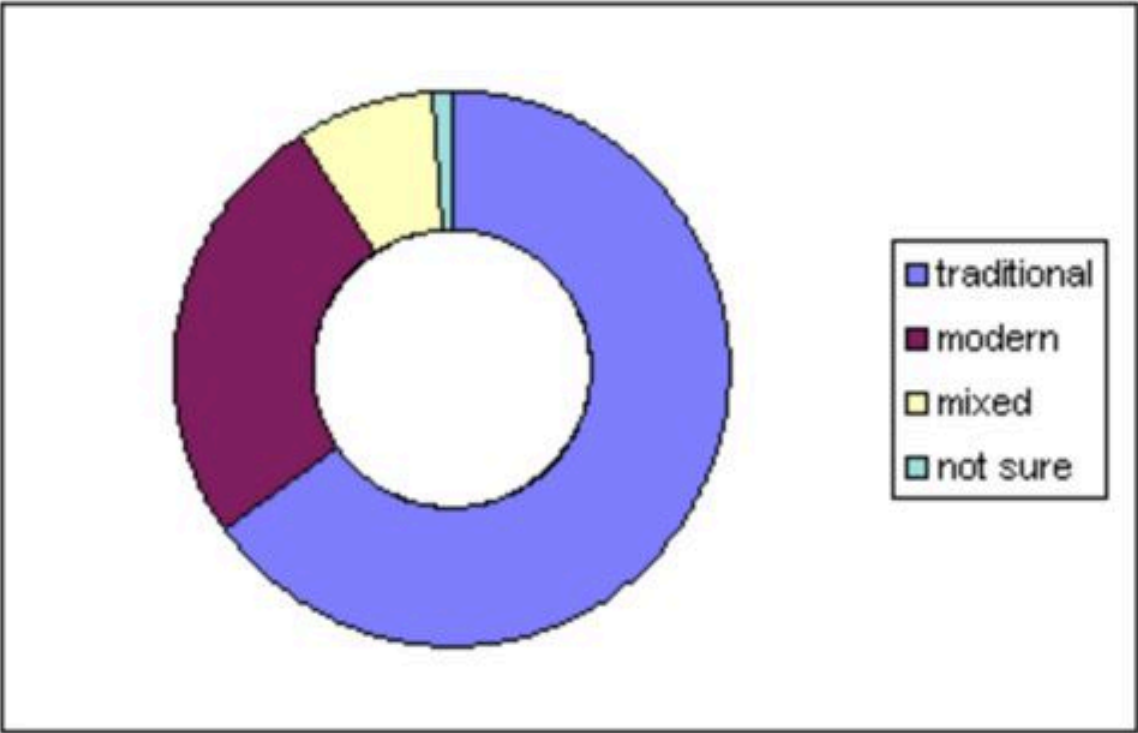
What is your favorite color?

Blue	44%
Green	12%
Red	11%
Black	4%
Purple/Violet	4%
Brown	3%
Pink/rose	3%
Beige/Tan	2%
White	2%
Grey	2%
Yellow	2%
Mauve	2%
Fushia	2%
Maroon	2%



When you select pictures, photographs, or other pieces of art for your home, do you find you lean more toward modern or more toward traditional styles?

traditional	64%
modern	25%
mixed	8%
not sure	1%

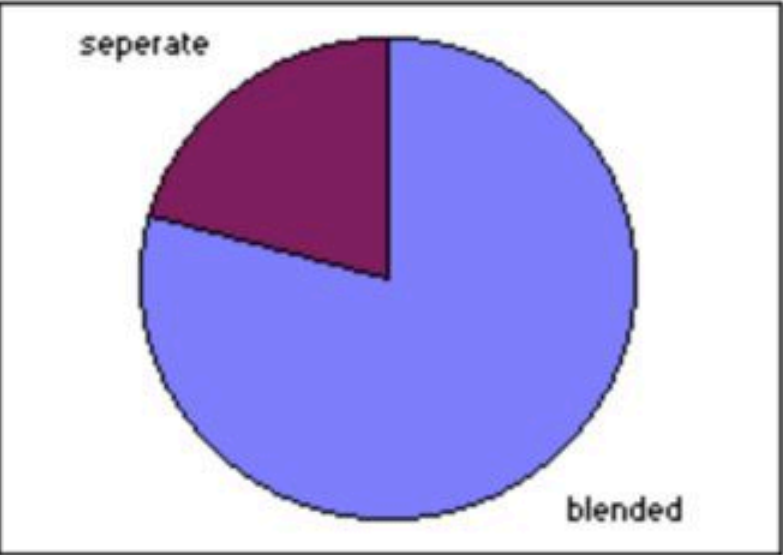


[All Survey Results](#) | [Homepage \(start over\)](#)

The United States survey was sponsored by the Nation Institute. The survey of 1001 adults was conducted by Marttila & Kiley, Inc.

Do you like to see colors blend into each other or do you like it when different colors are kept apart?

blended	68%
seperate	18%



And from what you've seen, would you say that you prefer paintings in which the person or people are nude, partially clothed or fully clothed?

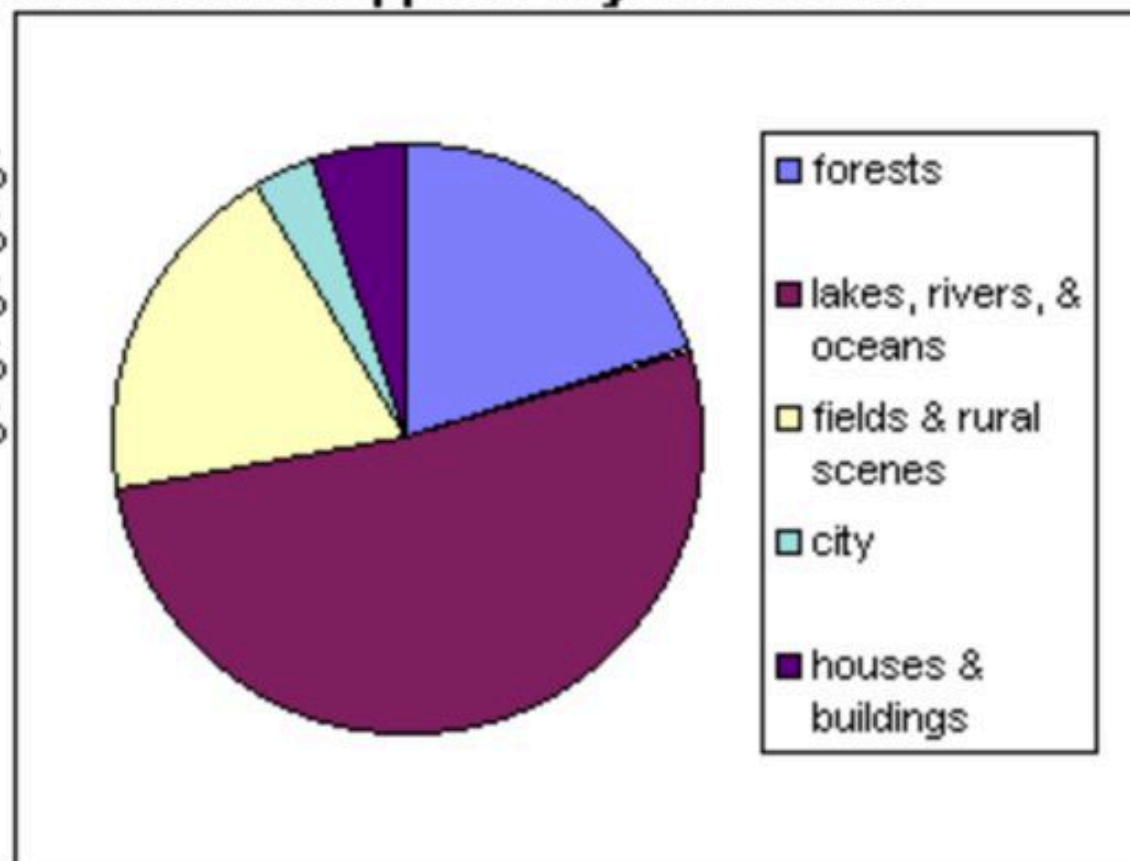
nude	3%
partially clothed	13%
fully clothed	68%



Which of the following outdoor scenes appeals to you the most?

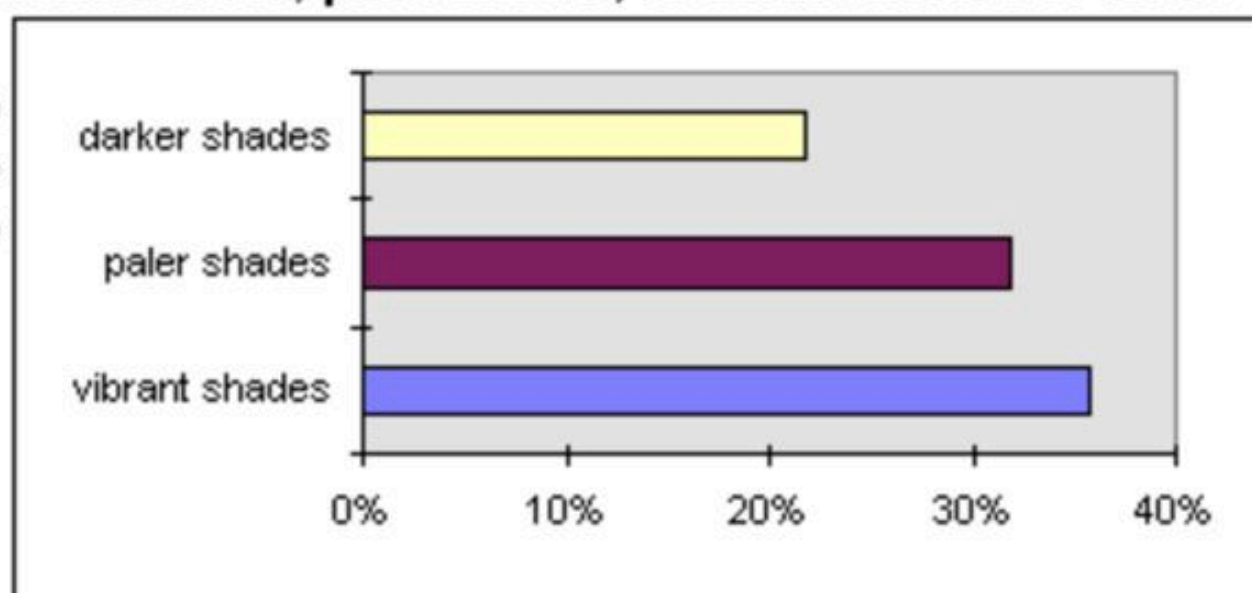
Paintings of:

forests	19%
lakes, rivers, & ocean	49%
fields & rural scenes	18%
city	3%
houses & buildings	5%



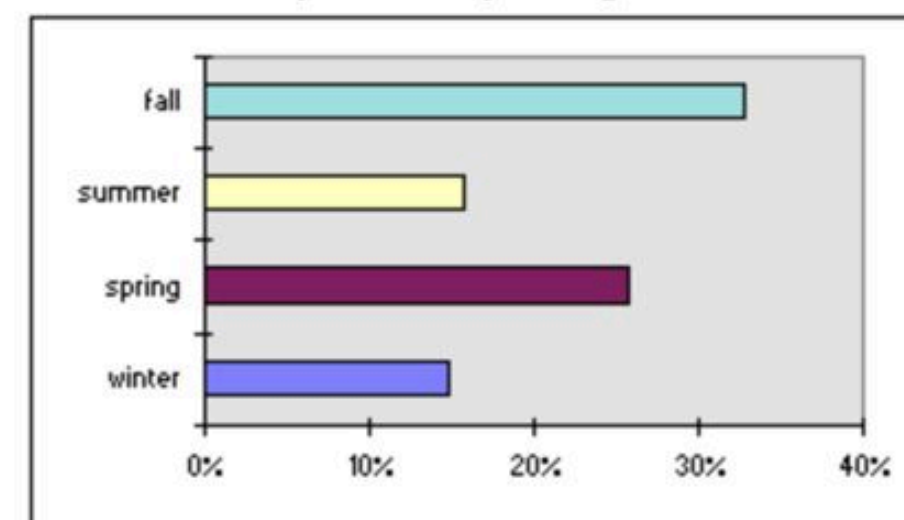
Every color has a wide range of shades. Take the color blue, for example. An artist can use a bright, vibrant shade of blue, like a royal blue, a pale, muted shade of blue, like light blue, or a darker tone of blue, like dark or midnight blue. In general, which would you say that you prefer? when the artist uses more vibrant shades, paler shades, or darker shades of color?

vibrant shades	36%
paler shades	32%
darker shades	22%



Which season would you most like to see depicted in a painting?

winter	15%
spring	26%
summer	16%
fall	33%





Komar & Melamid, *America's Most Wanted*, 1994



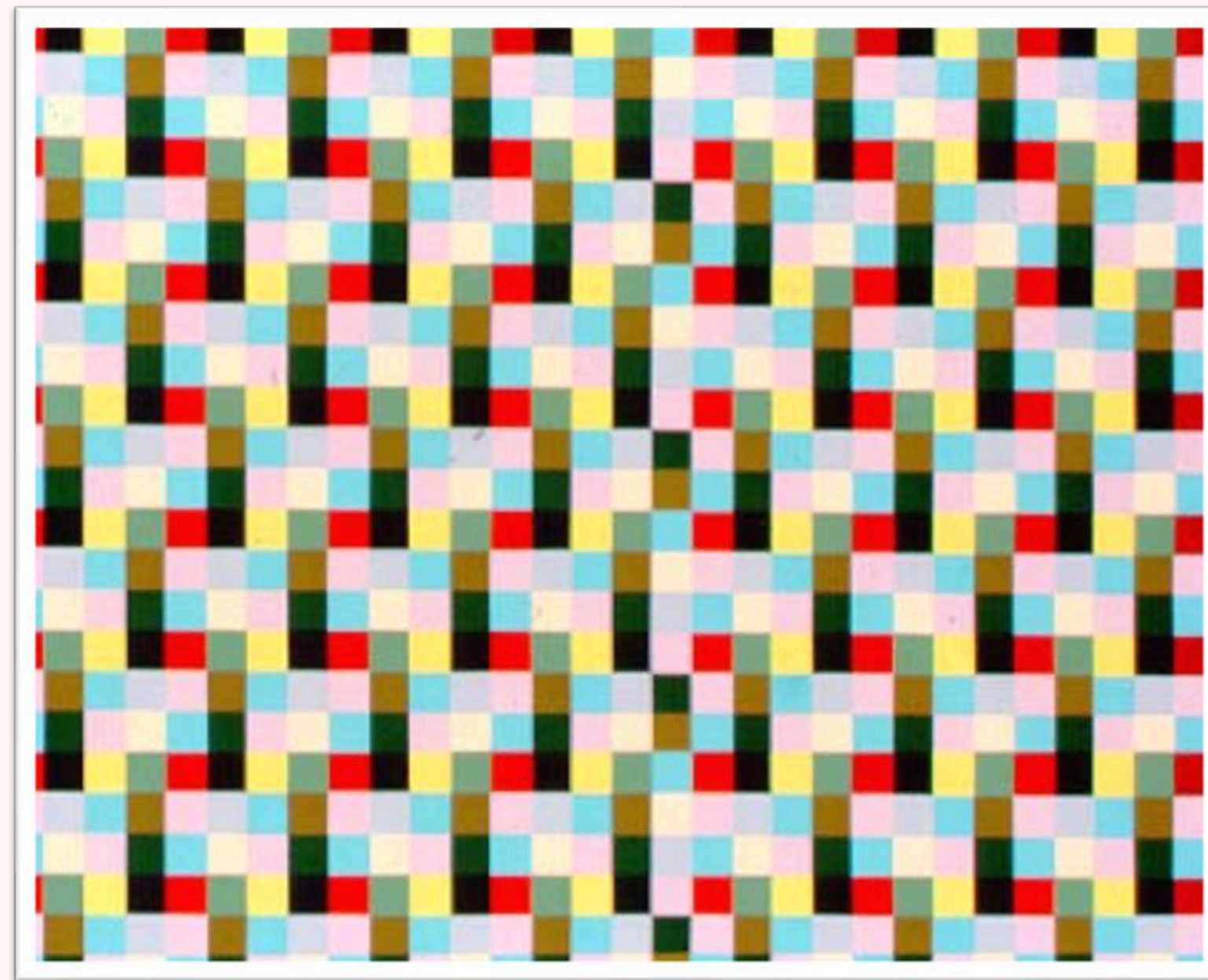
Komar & Melamid, *America's Least Wanted*, 1994



Komar & Melamid, *Iceland Most Wanted*, 1994

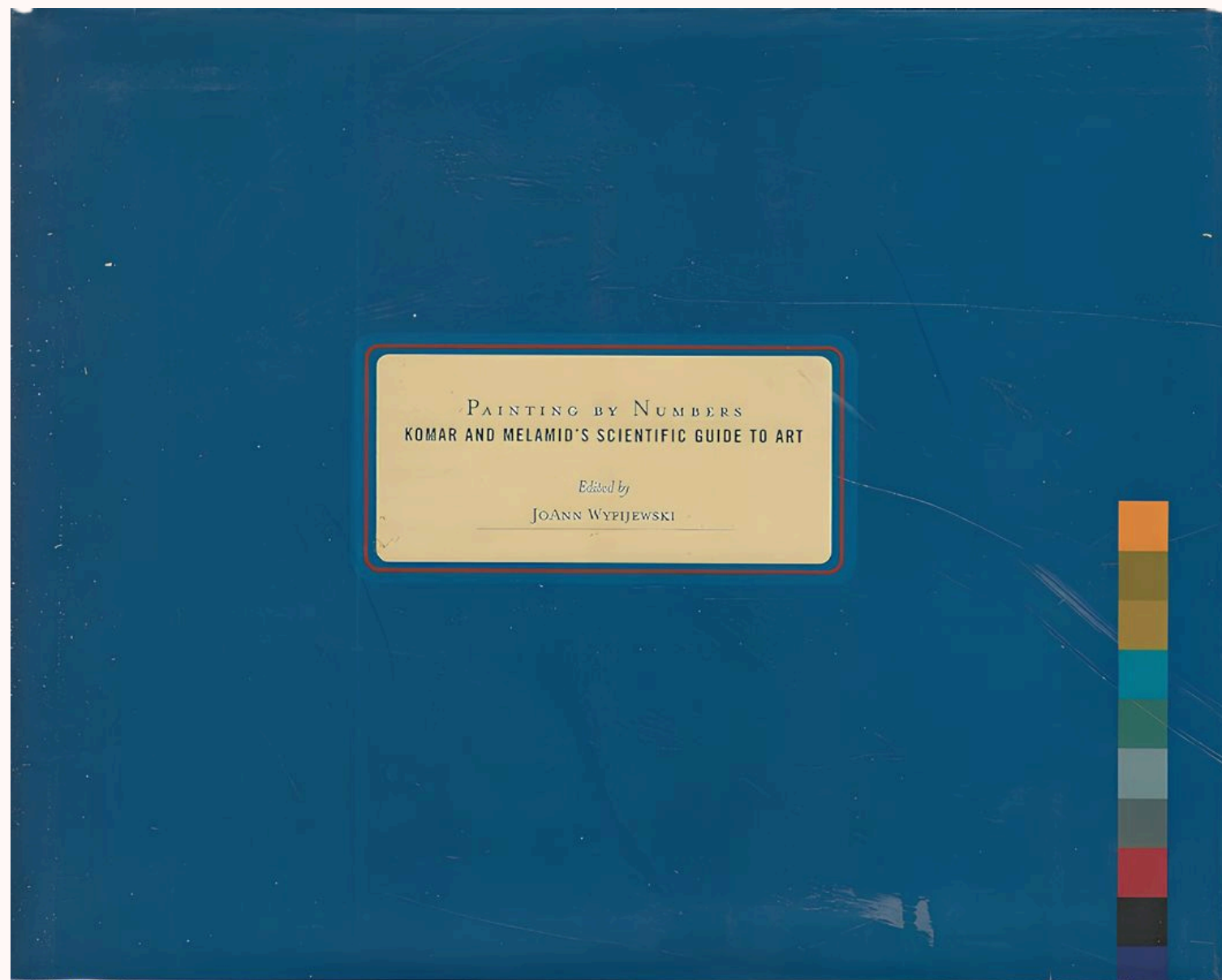


Komar & Melamid, *Germany Most Wanted*, 1994



Komar & Melamid, *Germany Least Wanted*, 1994

The People's Choice



Overview

In 1994, artists **Komar and Melamid** created paintings based on public opinion.

Insight

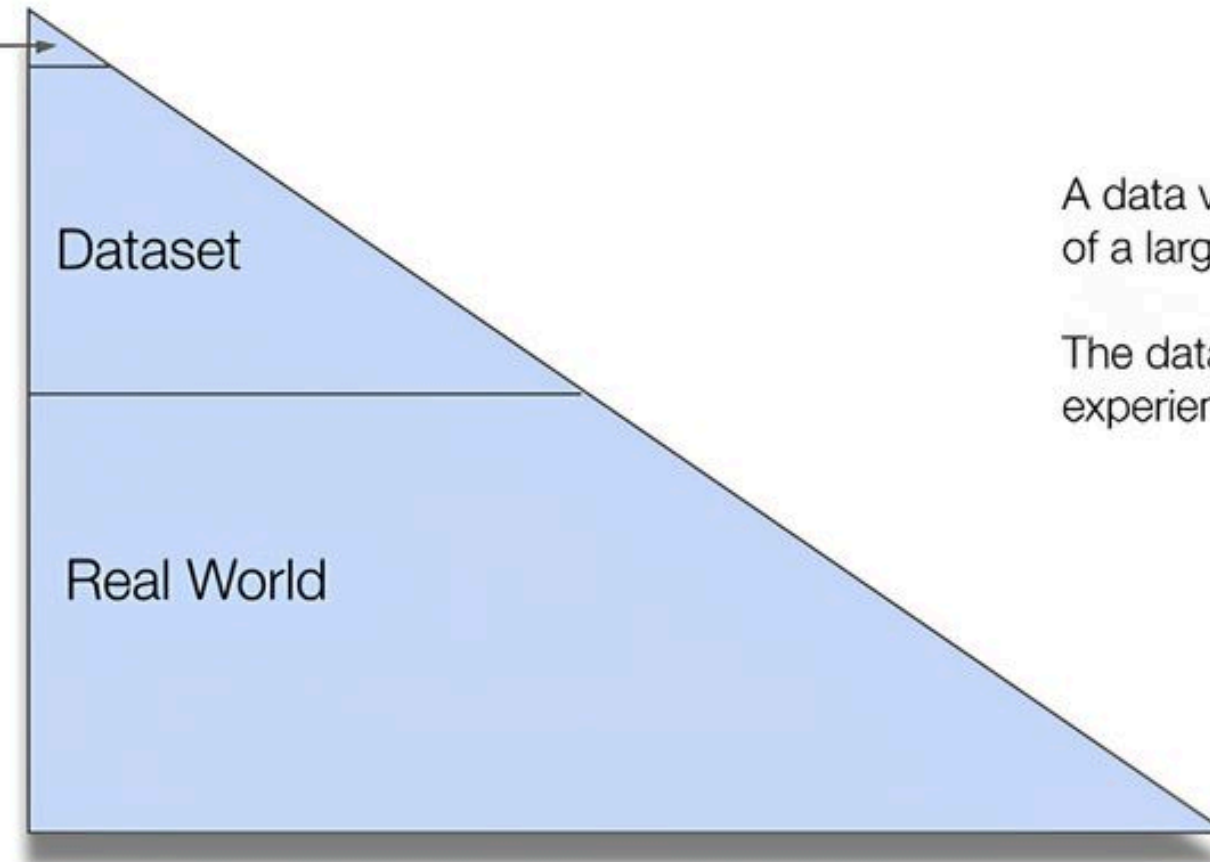
The results showed a **clear trend**: audiences favored a blue landscape featuring water and trees.

Komar, Vitaly, and Aleksandr Melamid. *Painting by Numbers: Komar and Melamid's Scientific Guide to Art*. Edited by JoAnn Wypijewski. New York: Farrar, Straus and Giroux, 1997

The “creativity” of machine learning is limited to the detection of styles from the training data and then random improvisation within these styles. In other words, machine learning can explore and improvise only within the logical boundaries that are set by the **training data**. For all these issues, and its degree of information compression, it would be more accurate to term machine learning art as **statistical art**

Pasquinelli, Matteo, and Vladan Joler. “The Nooscope Manifested: AI as Instrument of Knowledge Extractivism.”
AI & Society 36, no. 4 (2021): 1263–1280. <https://doi.org/10.1007/s00146-020-01097-6>.

Data
Visualization

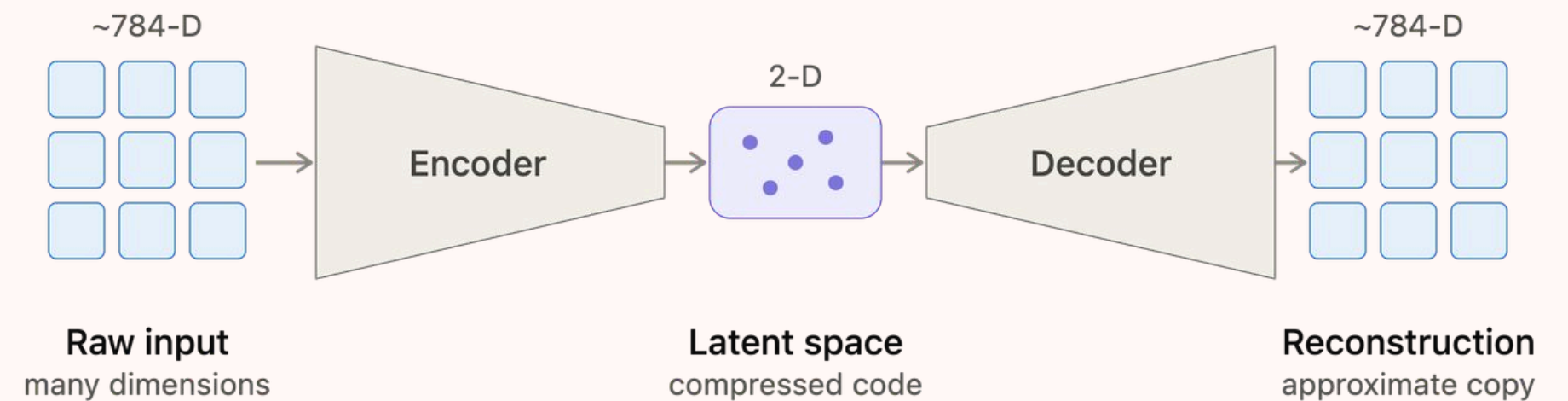


A data visualization represents a small portion of a larger dataset.

The dataset represents a small portion of the experienced world.

Latent space

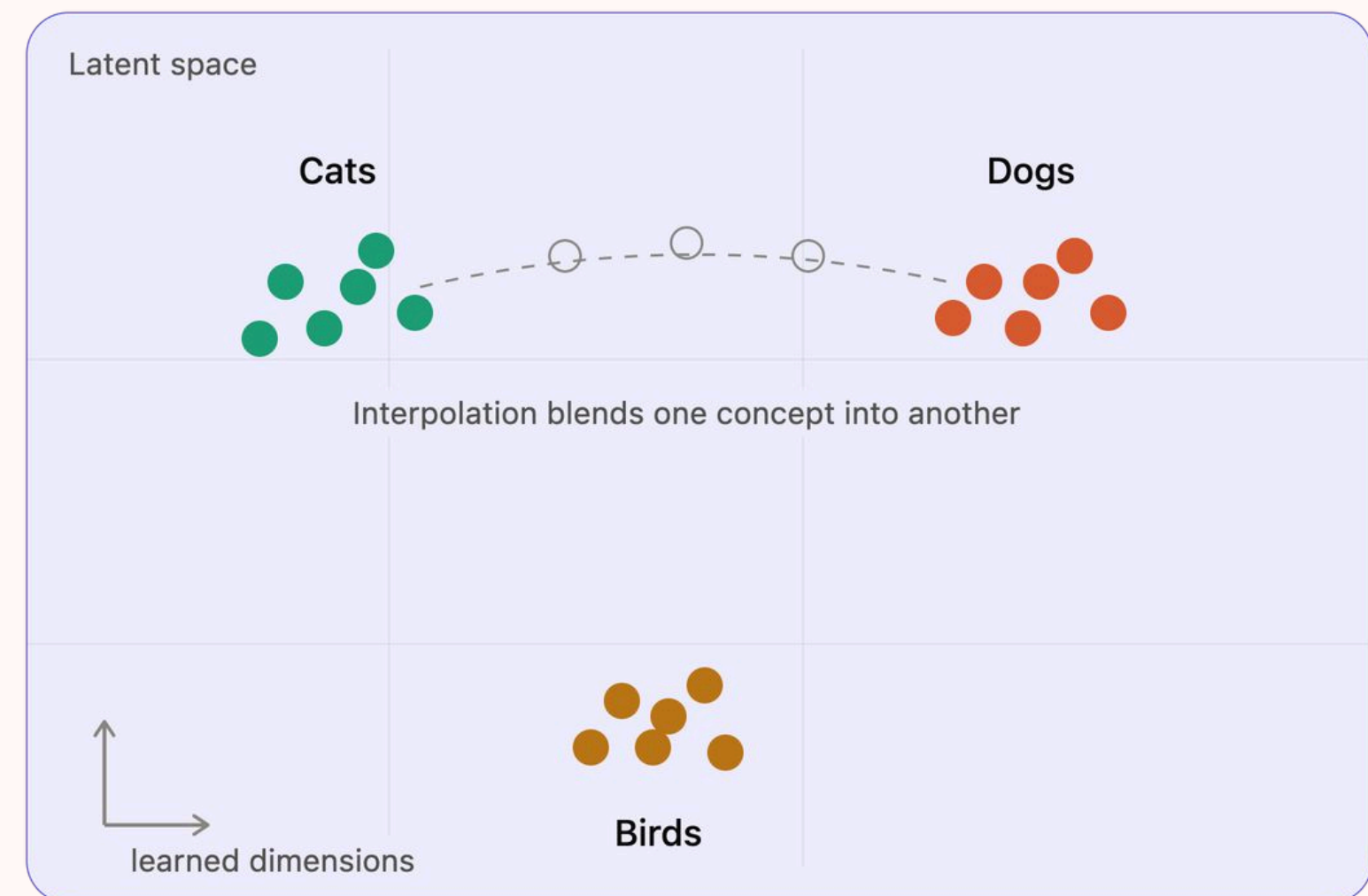
The latent space is the model's internal representation of meaning. It is a mathematical space learned during training, where similar concepts are positioned close to one another and dissimilar concepts are placed farther apart.



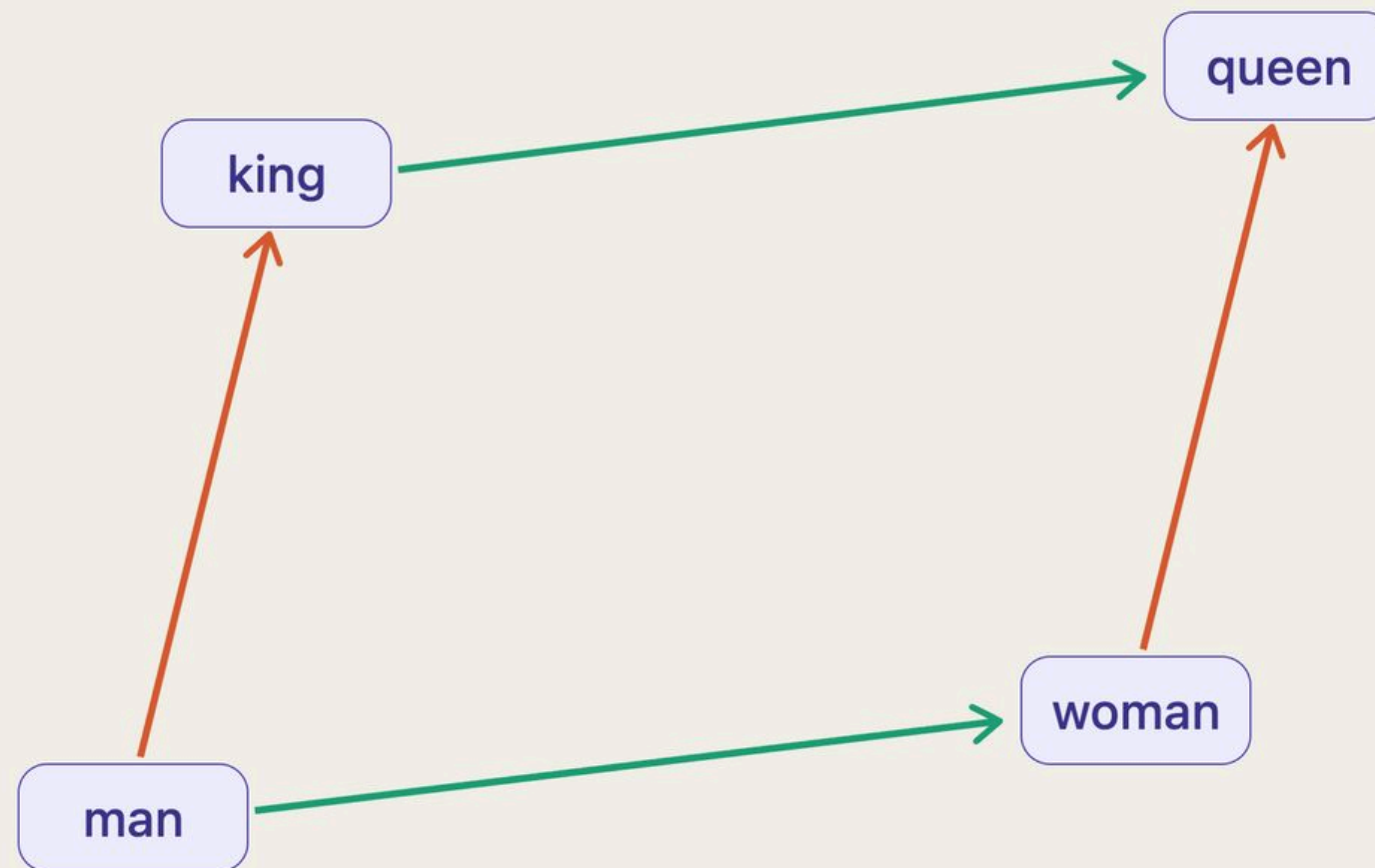
Proximity means similarity. Because the model packs related things together, distance in latent space is a measure of "how alike are these?"

The space is continuous. You can walk a straight line from one point to another and pass through valid in-between states.

The axes end up encoding **meaningful, learned concepts** rather than raw pixels.



$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$



— gender direction (male → female)

— royalty direction (person → royal)

When words get mapped to vectors, the model **lines up relationships as consistent directions**. So the offset that turns "man" into "woman" is roughly the same offset that turns "king" into "queen."

The Blue Landscape at Industrial Scale

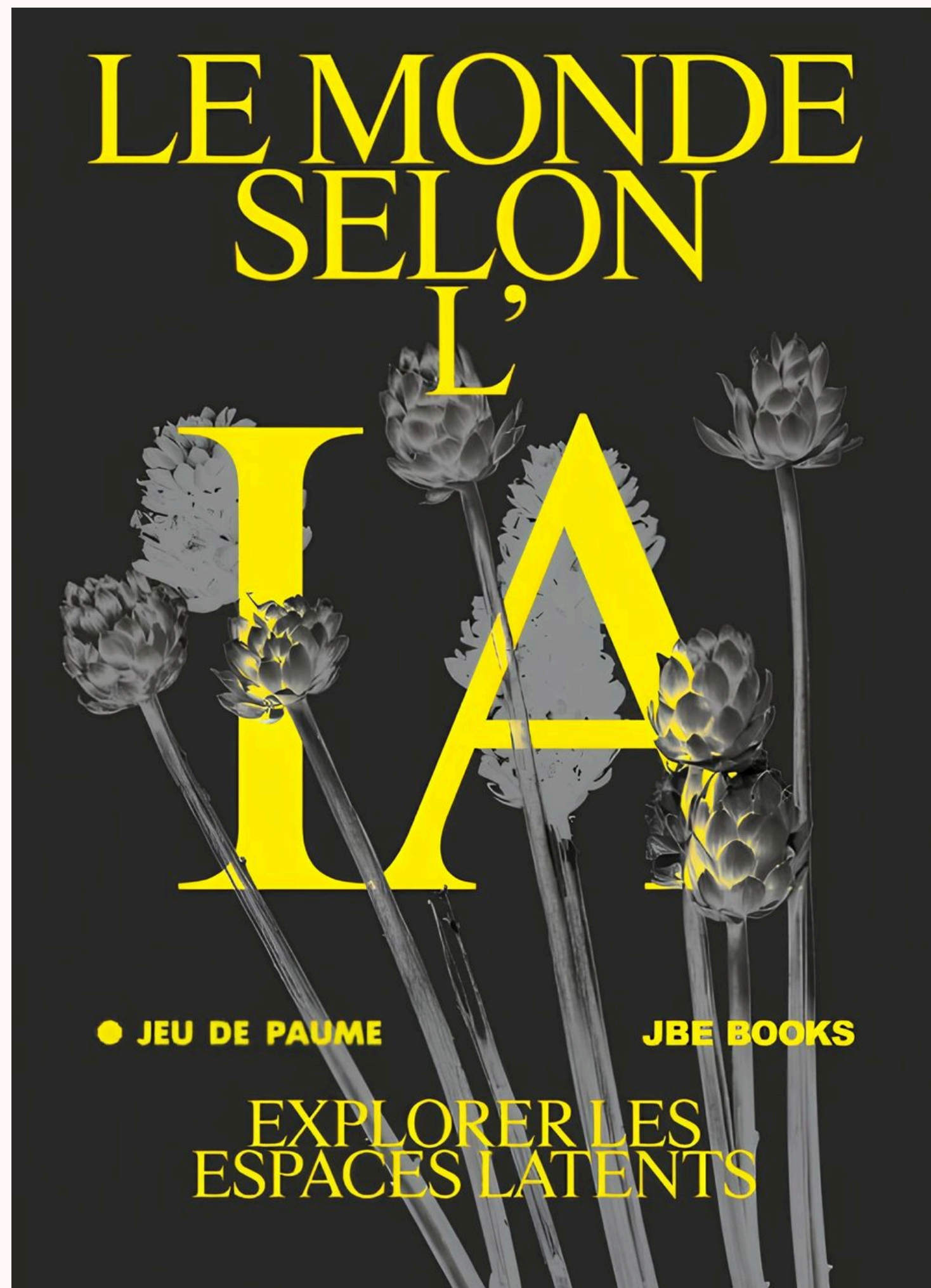
A **census**. No precise “author”, no referent — a distribution.

Human feedback: the “AI aesthetic”. Polished, over-lit, vaguely cinematic, interchangeable.



But...

Latent space is not an average, it also contains the improbable and the monstrous.



Le Monde selon l'IA, exhibition, curated by Antonio Somaini, Jeu de Paume, Paris, April 11–September 21, 2025.

Holly Herndon and Mat Dryhurst, *xhairymutantx* (2024)



Your image of Holly Herndon is ready.



"Holly Herndon" is already a token: her features surface in any image model.

The response: retrain the model on distorted, costumed images of Holly.

Mutant versions of the artist: self-inflicted data poisoning.



**xhairymutantx began by asking the question:
do we get to choose how we are represented
on the AI substrate?**



Describe something



Try something like: cyberpunk snail in a cloud

100/100

Trevor Paglen, *Adversarially Evolved Hallucinations* (2017)



Vampire (Corpus: Monsters of Capitalism)



Highway of Death (Corpus: The Aftermath of the First Smart War)

Antonio Somaini: «AI and Visual Culture: A Theory of Latent Spaces»



A theory of images and visual culture, today, needs a theory of latent spaces. In a historical phase in which images are more and more generated, modified, circulated, seen and described by or with the help of different kinds of AI models, we need to understand the crucial role played by an abstract, mathematical construct whose cultural and political implications could hardly be overestimated.

Latent spaces play also a central role in the contemporary artistic practices that engage with AI: whether to critically respond to its increasing presence in every aspect of culture, society, politics and economics, or to use it as a series of new tools for artistic production. For a few years now, artists have developed different strategies to explore or modify the existing, dominant latent spaces, or to produce their own alternative, antagonist, counter-hegemonic ones. Taken together, these different strategies show the awareness with which the field of contemporary art is tackling the presence and the agency of this hidden layer of mathematical abstraction that is profoundly transforming the status of images and vision, as well as the relations between images and other media.

Latent space is a new cultural infrastructure, reshaping visual culture much as the archive, library, and museum once did.

AI-generated images redefine key visual concepts, replacing direct referents with statistical ones and turning style into a transferable computational feature.

Latent space is a generative archive.

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